30 October 1959
SUPERSEDING
NAVORD OS 8129B
20 July 1959

MILITARY SPECIFICATION

SENSITIVITY SWITCH MARK 5 MOD 1

1. SCOPE

1.1 This specification describes the requirements for the procurement of the Sensitivity Switch Mark 5 Mod 1, BUORD LD 139391.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications or such portions thereof as designated herein of the issue in effect on the date of invitation for bids, or as noted, form a part of this specification.

SPECIFICATIONS

FEDERAL

PPP-T-60	Tape; pressure sensitive adhesive, waterproof-for packaging and sealing
PPP-B-676	Boxes, set-up, paperboard
PPP-C-843	Cushioning material, cellulosic
PPP-B-636	Boxes, fiber

MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-131	Barrier material; water vapor-proof, flexible
MIL-C-442	Cable, (wire), two-conductor, parallel, (ripcord)
MIL-S-6872	Soldering process, general specifications for

FSC 5930

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STANDARDS

MILITARY

MIL-STD-105	Sampling procedures and tables for inspection by attributes
MIL-STD-129	Marking for shipment and storage

DRAWINGS

NAVY DEPARTMENT

BUORD LD 139391	Sensitivity Switch Mark 5 Mod 1
BUORD LD 274820	Test Set Mark 126 Mod 0

And all documents listed in the above LD's and drawings.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400	General instructions for design, manufacture and inspection of Naval Ordnance Equipment

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 and all supplements thereto	Screw-thread standards for federal services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.).

3. REQUIREMENTS

3.1 Description.- The Sensitivity Switch Mark 5 Mod 1 is a manually operated five position attenuator used to reduce the sensitivity of underwater mines.

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3.2 General requirements.

3.2.1 Preproduction samples. - Unless otherwise specified in contract or order, preproduction samples of the Sensitivity Switch Mark 5 Mod 1 shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification, and the design. See paragraph 4.1.2.

3.2.2 Materials. - Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the Sensitivity Switch Mark 5 Mod 1, shall conform to the requirements indicated on the documents listed in BUORD LD 139391.

3.2.3 Finishes, coatings, construction and dimensions. - The finishes, coatings, construction, and dimensions of the Sensitivity Switch Mark 5 Mod 1 shall be in accordance with the documents listed in BUORD LD 139391.

3.2.4 Moisture, fungus and corrosion. - The Sensitivity Switch Mark 5 Mod 1 shall be rendered moisture, fungus and corrosion resistant in accordance with detail requirements specified on the drawings and specifications in BUORD LD 139391.

3.2.5 Marking. - Marking of assemblies, sub-assemblies, and parts shall be in accordance with the applicable documents listed in BUORD LD 139391.

3.2.6 Interchangeability. - Component parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in strict agreement with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required then changes in dimensions are necessary and revision and correction of the applicable drawings are required. See 3.2.10.

3.2.7 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the

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contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.8 Threads. - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28 and supplements thereto. The class or fit for threads shall be as specified on the applicable drawings, and the NOT-GO gaging practice shall be as set forth in the supplement to the aforementioned Handbook H28.

3.2.9 Requests for deviation. - Requests for deviation from this specification or any document pertaining to this specification shall be as set forth in OP 400.

3.2.10 Conflicting requirements. - Conflicting requirements arising between this specification and any other specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.2.11 The requirements and procedures established in OP 400 shall be complied with, as applicable.

3.3 Performance requirements and product characteristics. - The Sensitivity Switch Mark 5 Mod 1 shall meet the following performance requirements.

3.3.1 Capacitance and dissipation factor. - When measured at 1000 cycles the capacitance of capacitor C-1 shall be 1.0 ± 0.2 microfarads and the dissipation factor shall be less than 0.08 (8 percent). Testin shall be as specified in 4.5.4.1.

3.3.2 Insulation resistance. - The insulation resistance between the cable leads and also between each cable lead and the chassis shall not be less than 50 megohms when tested at 500 volts d-c. Testing shall be as specified in 4.5.4.2.

3.3.3 Assembly. - The sensitivity switch shall pass all tests performed by Test Set Mark 126 Mod 0 as specified in 4.5.4.3.

3.4 Environmental requirements.

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3.4.1 Vibration. - The Sensitivity Switch, Mark 5 Mod 1, shall meet all performance requirements and product characteristics of paragraph 3.4 after being mounted and tested as specified in paragraph 4.6.1.

3.4.2 Shock. - The Sensitivity Switch Mark 5 Mod 1 shall meet all performance requirements and product characteristics of paragraph 3.4 after being tested as specified in paragraph 4.6.2.

3.4.3 Visual inspection (unpotted). - Units designated as preproduction samples or periodic production samples shall be unpotted and visually inspected for compliance with the requirements of the drawings with respect to the location and type of component parts and wiring methods. The test for this requirement shall be considered a destructive test and shall be performed only after all other inspections have been completed. See 4.6.3.

3.4.4 Low Temperature Flexibility Test. - The Cable Assembly CA-617 shall withstand the temperature conditions specified in 4.6.4 and thereafter show no corrosion or damage and shall meet the performance requirements and product characteristics of this specification.

3.5 Clean-up. - Prior to and after final assembly, all parts, components and the assembly shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.6 Workmanship. - The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting machine screw assemblies, welding, brazing and freedom of parts from burrs and sharp edges.

3.6.1 Soldering. - Unless otherwise specified, soldering processes shall conform to the requirements specified in Specification MIL-S-6872. Electrical connections shall be mechanically secure before soldering wherever possible. Only rosin flux containing no acid shall be used. The use of rosin-core solder is permitted.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Government inspection. - The Government inspector will make such inspections as are necessary to determine that components and assemblies are in accordance with the requirements of the contract, pertinent drawings and specifications. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in OP 400.

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4.1.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection, shall be in accordance with the provisions set forth in MIL-STD-105.

4.1.2 Preproduction sample. - Unless otherwise specified for each contract or order, a representative sample of twenty (20) sensitivity switches shall be manufactured using the methods and procedures proposed for the production lot. These shall be submitted to the Government inspector for preproduction tests at an activity designated in the contract or by the Bureau of Ordnance for purposes of determining compliance with the requirements of the contract, specifications and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on the preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.3 Lot size. - As applied to Government inspection of units of product the term "lot" shall mean "inspection lot", i.e., a collection of units of product submitted by a supplier for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" will be as determined by the Government inspector and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose

4.1.4 Periodic production samples. - A sample of sensitivity switches shall be randomly selected by the Government inspector from each lot in accordance with MIL-STD-105, Appendix Inspection Level L-7. The sensitivity switches shall be considered satisfactory if the sensitivity switches meet all specifications and pertinent drawing requirements and withstands the tests as specified in 4.5 and 4.6 inclusive.

4.2 Contractor's inspection. - Contractor's inspection shall be conducted in accordance with provisions of "Inspection by Contractor" as contained in OP 400.

4.3 Classification of tests. - The inspection and testing of the sensitivity switches shall be classified as follows:

4.3.1 Acceptance tests. - These tests are to be accomplished on each sensitivity switch being submitted for acceptance under contract. Acceptance tests shall be performed by the manufacturer and witnessed by the Government inspector. These tests are detailed in 4.5.

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4.3.2 Preproduction and periodic production tests. - Preproduction and periodic production tests are those which are accomplished on samples selected as specified in 4.1.2 and 4.1.4 which are representative of the production of the sensitivity switches after the award of the contract to determine that the production meet the requirements of this specification. These are detailed in 4.6. Acceptance shall be based on no defects in the sample. Failure of the sample to comply with these requirements will result in the rejection of the lot, or the cessation of production as determined by the procuring activity.

4.4 Test equipment.

4.4.1 Test and inspection equipment and facilities. - Unless otherwise specified, the manufacturer shallfurnish and maintain all necessary test equipment, facilities, and personnel for performing all acceptance tests under the supervision of the Government inspector. The test equipment shall be adequate in quantity, and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.4.2 The following items of test equipment are required to perform the acceptance tests set forth in this specification:

4.4.2.1 Test Set Mark 126 Mod 0, minimum of two (2) required. (To be furnished by the Government).

4.4.2.2 Megohmmeter capable of measuring 50 megohms at 500 volts d-c, accuracy to be within ± 5 percent.

4.4.2.3 Bridge circuit capable of measuring, at 1000 cycles, a capacitance of 1.0 ± 0.2 microfarads, accuracy within ± 2 percent, and a capacitor dissipation factor of 0.08, accuracy within ± 20 percent.

4.5 Acceptance tests.

4.5.1 The contractor or Government inspector shall conduct acceptance tests as specified in 4.3.1 to assure that Sensitivity Switch Mark 5 Mod 1 is in compliance with the requirements of 3.2, 3.3, 3.5 and 3.6.

4.5.2 Test conditions. - Unless otherwise specified the Sensitivity Switch Mark 5 Mod 1 shall be subjected to acceptance tests under the following conditions:

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4.5.2.1 Temperature: Room ambient 65°F to 95°F (18°C to 35°C).

4.5.2.2 Altitude: Normal ground atmospheric pressure.

4.5.2.3 Vibration: None.

4.5.2.4 Humidity: Room ambient to 95° relative maximum.

4.5.3 Visual inspection. - The sensitivity switch and its component parts shall be inspected to verify that the materials, design, construction, physical dimensions, soldering, and workmanship are in accordance with this specification and the drawings.

4.5.4 Test procedure.

4.5.4.1 Capacitance and dissipation factor. - In order to test the capacitor in the sensitivity switch, first remove the cap to allow access to pin G. Connect the bridge circuit, 4.4.2.3, across pin G in the receptacle and the terminal on wire J of cable assembly CA-617 and measure the capacitance and dissipation factor to determine compliance with the requirements of 3.3.1.

4.5.4.2 Insulation resistance. - Remove the cap from the sensitivity switch and, using megohmmeter, 4.4.2.2, measure the insulation resistance between the cable leads and each of the two cable leads and the chassis to determine compliance with the requirements of 3.3.2.

4.5.4.3 Assembly. - Test the assembled sensitivity switch with Test Set Mark 126 Mod 0 (item 4.4.2.1) as specified in the operating instructions supplied with the test set.

4.6 Preproduction and periodic production tests.

4.6.1 Vibration. - The Sensitivity Switch Mark 5 Mod 1 shall be rigidly fastened directly to the vibration table by either its normal mounting attachments or by its external case whichever is appropriate. Simple harmonic excitation shall be applied parallel to each of the three principal axes of the device for equal periods of time. The frequency range from 10 to 60 cps may be covered by slowly cycling at a logarithmic rate between the limits or by at least 24 discrete frequency steps which have a logarithmic distribution. The vibration amplitude shall be maintained at 2± 0.20 vector or 0.10 ± 0.01 inches (peak-to-peak) whichever is the lesser. The duration of the test in each direction is eight (8) hours. The total test duration shall be twenty-four (24) hours.

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4.6.2 Shock. - The Sensitivity Switch shall be subjected to a continuous two phase shock applied in any two mutually perpendicular directions.

(a) Impact phase. - This phase shall consist of a minimum change in velocity of 18-20 feet per second taking place in 0.20 to 0.40 millisecond.

(b) Drag phase. - This phase shall subject the Sensitivity Switch to an acceleration for a minimum of 15 milliseconds in such a manner that the average acceleration is 350 gravity units. The peak acceleration shall be between 500 and 550 gravity units.

4.6.3 Visual inspection (unpotted). - The Sensitivity Switch Mark 5 Mod 1, after having been inspected for all other requirements of the drawings and this specification, shall be unpotted by melting the encapsulating compound. The unit shall then be inspected for compliance with 3.4.3.

4.6.4 Low Temperature Flexibility Test. - The Cable Assembly CA-617 shall be subjected to temperature tests as specified in 4.6.4.1 and thereafter shall meet the performance requirements and product characteristics of this specification.

4.6.4.1 The Cable Assembly CA-617 shall be subjected to Method 4.7.6, Low Temperature Flexibility Test of MIL-C-442.

4.7 Acceptance criteria. - Sample units which fail to comply with any of the test or inspection requirements shall be classified as defective. If the number of defective units exceeds the number of defectives permitted by the associated acceptance criteria the lot represented will be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A.

5.1.1.1 Prior to unit packaging, Cable Assembly CA-617, shall be coiled and secured with tape, Federal Specification PPP-T-60, type and grade optional.

5.1.1.2 Each Sensitivity Switch Mark 5 Mod 1, prepared as specified above, shall be packaged and cushioned in a Type I paperboard set-up box conforming to the requirements of Federal Specification PPP-B-676. A one (1) inch thickness of cellulose wadding, Type III conforming to the requirements of Federal Specification PPP-C-843, shall be utilized within

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the unit container to cushion switch assembly.

5.1.1.3 Each Sensitivity Switch Mark 5 Mod 1, individually packaged as described above, shall be further packaged Method IA-15, conforming to the requirements of Military Specification MIL-P-116.

5.1.1.3.1 The bag shall be fabricated from a class 2 flexible water vaporproof barrier-material conforming to the requirements of Military Specification MIL-B-131.

5.1.1.4 Ten (10) Sensitivity Switches Mark 5 Mod 1, individually packaged as specified above, shall be intermediately packaged in a Style CSSC fiberboard box conforming to the requirements of Federal Specification PPP-B-636, Class 2, compliance symbol W5c or W5s.

5.1.2 Level C

5.1.2.1 Sensitivity switches shall be packaged in a manner which will afford adequate protection from deterioration and physical damage during shipment from the contractor or supply source to the first receiving activity for immediate use.

5.2 Packing

5.2.1 Level A

5.2.1.1 Four (4) intermediate containers containing a total of forty (40) switch assemblies prepared as specified above for Level A protection, shall be packed in an overseas type wood-cleated plywood box or in a Class 2, style 2, 2 1/2, 3 or 4 nailed wood box conforming to the requirements of Federal Specification PPP-B-636. Sufficient fiberboard fillers fabricated from Class 2, W5c or W5s material conforming to Federal Specification PPP-B-636, shall be utilized within the shipping container to provide a tight pack.

5.2.2 Level C

5.2.2.1 Sensitivity switches shall be packed in a manner which will provide protection from damage during direct shipment from the contractor or supply source to the first receiving activity for immediate use. The type of containers used shall conform to applicable carrier rules and regulations.

5.3 Marking

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5.3.1 The following identification marking shall be applied to unit packages with labels:

STOCK NO.
(1) SENSITIVITY SWITCH MARK 5 MOD 1
DATE PKD METHOD OF PRES IA-15, MIL-P-116
SER. NO. CONTRACT NO.
MANUFACTURER'S NAME AND ADDRESS

5.3.2 The following identification marking shall be stenciled or stamped with 1/4" (min) high characters on the top and one (1) side of the intermediate container:

STOCK NO.
(10) SENSITIVITY SWITCH MARK 5 MOD 1
DATE PKD LOT NO.
CONTRACT NO.
MANUFACTURER'S NAME AND ADDRESS

5.3.3 The following identification marking shall be stenciled or stamped with 1/2" (min) high characters on the upper two-thirds (2/3) of one (1) end and the upper one-third (1/3) of one side (to the right when facing the marked end) of the shipping container:

STOCK NO.
(40) SENSITIVITY SWITCH MARK 5 MOD 1
DATE PKD LOT NO.
GROSS WT. LBS. CUBE CU. FT.

5.3.3.1 The following contract data marking shall be stenciled or stamped with 1/2" (min) high characters on the right upper one-third (1/3) of the opposite side (to the left when facing the marked end) of the shipping container:

NAME AND ADDRESS OF CONTRACTOR
CONTRACT NO.

5.3.4 In addition to any special marking required by the contract or order, all marking shall be in accordance with Military Standard MIL-STD-129.

6. NOTES

6.1 Intended use. - The Sensitivity Switch Mark 5 Mod 1 is a component of underwater mines.

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6.2 Ordering data. - Procurement documents should specify the desired level of packaging and packing. Level C should not be selected unless it is definitely known that the equipment will be used immediately upon receipt.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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